



PROJECT-BASED LEARNING

PHASE-I EVALUATION

SkillArc

Team ID: T028

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27



Team & Mentor Details

Team Information

Team ID: T028

Semester: 3rd

Section: ML (1, 2, 3)

Domain: EdTech

Team Members

1. Samiksha Gurung – 2510370150
2. Tanisha Rai – 2510370080
3. Vaibhavi Thapliyal – 2510370530

Mentor

Mr. Kuldeep Nautiyal

Interactions completed:3 _____

Problem Statement & Motivation

Problem Statement

Learning technical subjects is often fragmented across multiple platforms, leading to unstructured study habits, poor consistency, and limited access to quality peer resources.

Motivation

- ▶ **Scattered Resources:** Makes learning unstructured.
- ▶ **Lack of Consistency:** Reduces regular coding practice.
- ▶ **Limited Peer Sharing:** Hard to access quality notes.
- ▶ **Low Motivation:** Causes students to lose interest over time.

Target Users & Stakeholders

- ▶ **Primary:** College students learning technical subjects.
- ▶ **Secondary:** Mentors, faculty, and peer contributors.
- ▶ **Beneficiaries:** Students seeking structured learning.

Objectives

Project Objectives

1. Provide structured subject-wise learning roadmaps.
2. Centralize notes, resources, quizzes, and coding practice.
3. Encourage consistency through Credit Points and streaks.
4. Enable peer-to-peer note and resource sharing.
5. Conduct weekly contests and maintain leaderboards.
6. Track learning progress and student contributions.

Key Objective: Build an all-in-one ecosystem that combines structured roadmaps, peer resource sharing, and gamified progress tracking.

Proposed Solution & Approach

Proposed Approach

1. **Input / Data Collection:** Student details, subjects, quizzes, and activities.
2. **Processing / Algorithm:** Roadmaps, search, sorting, and credit calculations.
3. **System Logic:** Track progress and award Credit Points.
4. **Database / Storage:** Store student progress, quiz, and credit data.
5. **Output / Visualization:** Dashboard, leaderboard, and rewards.

Key System Modules

- ▶ Subject Roadmaps
- ▶ Centralized Notes & Quizzes
- ▶ Coding Practice Module
- ▶ Peer-to-Peer Sharing
- ▶ Credit Points & Streaks
- ▶ Dynamic Leaderboards

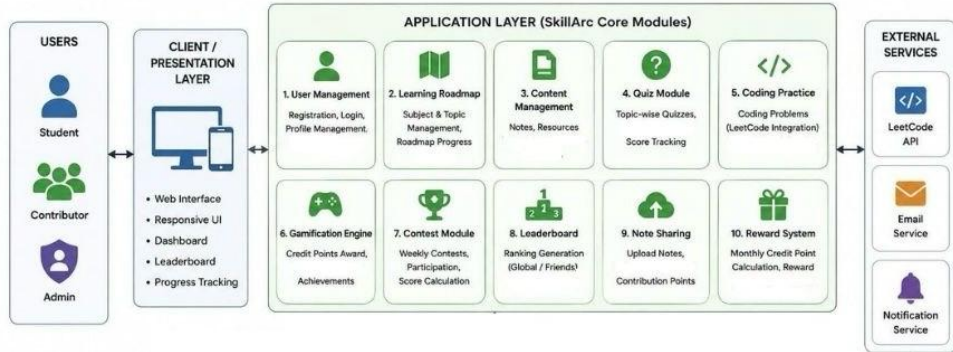
Integration of PBL Course Concepts

Course	Concepts Applied	Module Used in SkillArc
Data Structures in C	Arrays, Linked Lists, Stacks, Queues, Graphs, Hashing	Data management, progress tracking, quizzes, leaderboard and rewards
OOPs with C++	Classes, Objects, Encapsulation, Inheritance, Polymorphism, File Handling	Student, Subject, Topic, Quiz, Contest, Note and Reward modules

Meaningful Integration: Data Structures handle data organization and searching/sorting algorithms, while OOP constructs model system entities and domain logic.

System Architecture / Workflow

SkillArc – High Level System Architecture

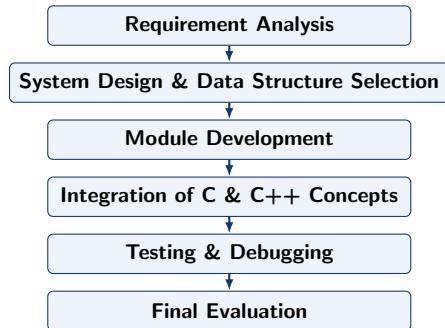


Technology Stack & Methodology

Technology Stack

- ▶ **Programming:** C / C++ / Python
- ▶ **Frontend Interface:** Streamlit
- ▶ **Storage:** File Handling / Database
- ▶ **Version Control:** Git / GitHub
- ▶ **IDE:** VS Code

Development Methodology



Initial Progress & Team Contribution

Progress Achieved

- ▶ Requirement analysis & domain research
- ▶ System architecture & workflow design
- ▶ Phase-I evaluation slides completed
- ▶ C++ basic concepts
- ▶ DSA basic topics

Individual Contribution

Member		Role	Contribution
Samiksha Gu-rung		Lead / Dev	System architecture, core logic & roadmap design
Tanisha Rai		Developer	Data structure implementation & quiz modules
Vaibhavi Thap-liyal		Developer/Docs	OOP integration, note module & testing setup

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Proposal & Design Phase
2. Phase-II: Core Development & Module Integration
3. Phase-III: Testing, UI Integration & Final Presentation

Expected Outcomes

- ▶ Functional platform for building programming skills.
- ▶ Interactive roadmaps, practice system & gamified credit points

References

1. Yashavant Kanetkar, Data Structures Through C, 5th Edition
2. Herbert Schildt, C++: The Complete Reference, 4th Edition, McGraw-Hill.
3. <https://roadmap.sh/roadmaps>
4. <https://www.geeksforgeeks.org/>
5. <https://www.youtube.com/>

Thank You
Questions & Suggestions